

Containers on Z

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What we will discuss

1. What is Z?
2. Why containers on Z?
3. Container options on Z.
4. Torch-Spyre: How we develop.
5. Demo AI workload zCX.



Red Hat



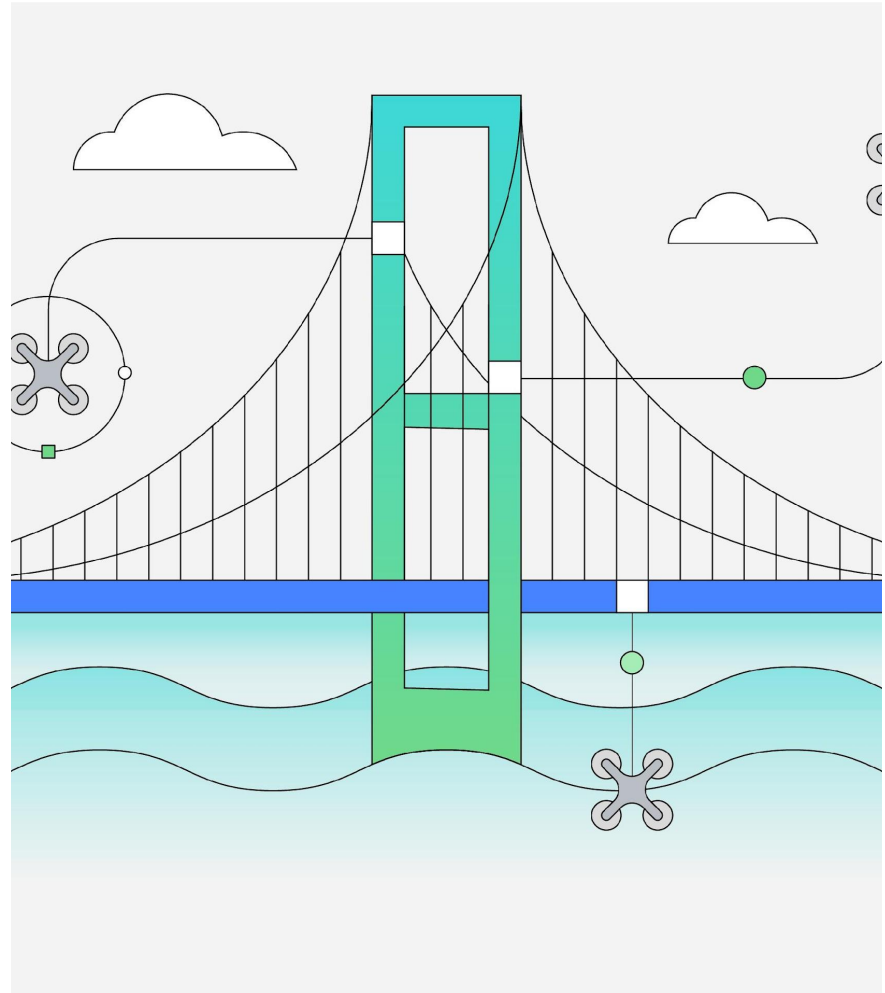
1. What is Z?

IBM Z = Mainframe

IBM Z = Zero Downtime

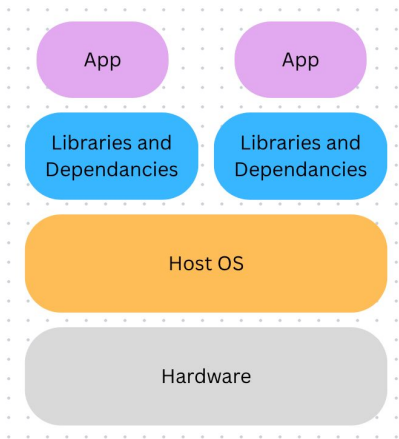
What runs on IBM Z:

1. Global banking transactions
2. Airlines reservation systems
3. Insurance & healthcare records
4. Government systems
5. Tax records, social security

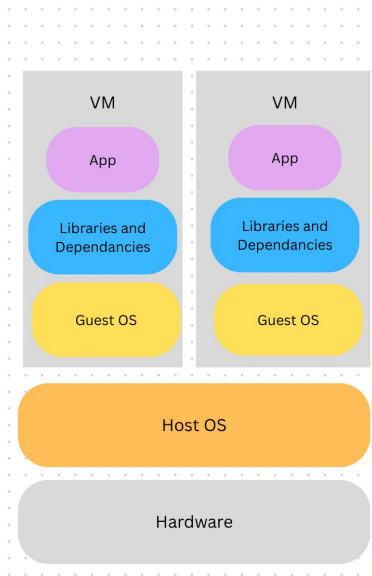


Evolution of Deployment

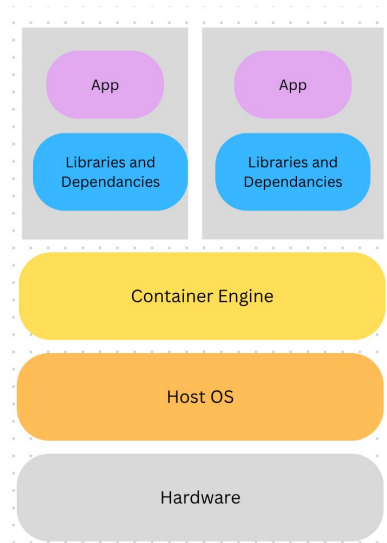
Personal Computer



Virtual Machines

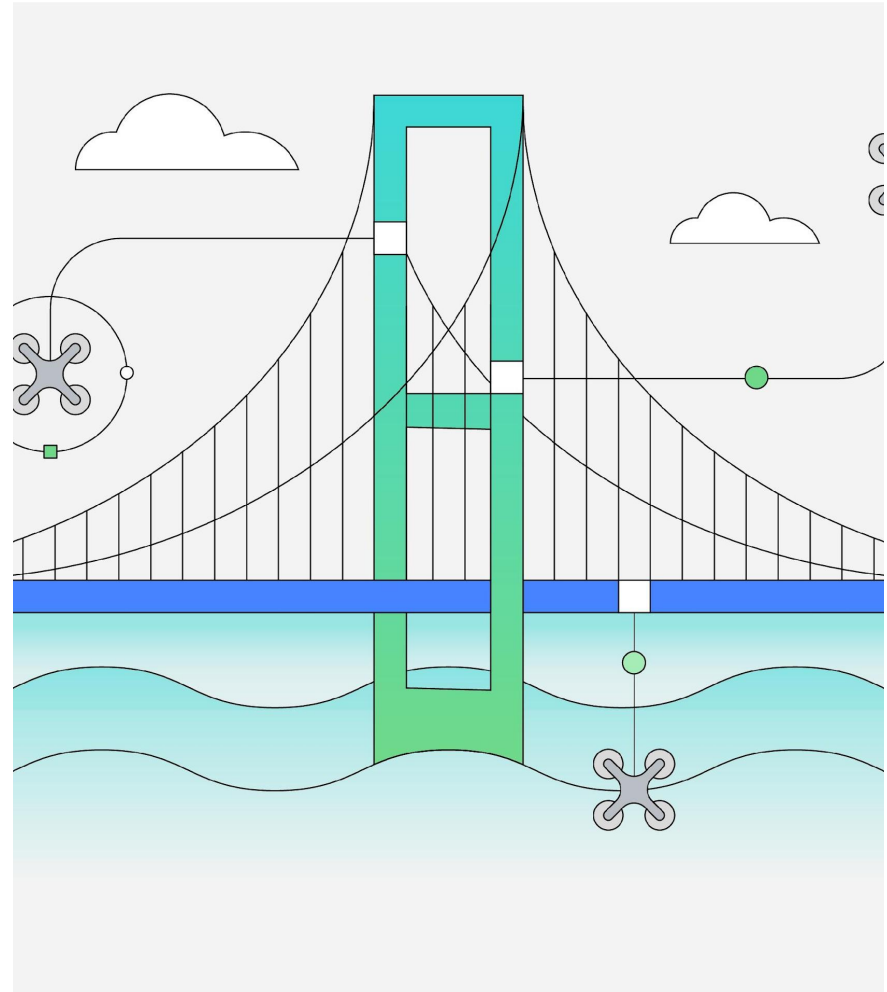


Containers

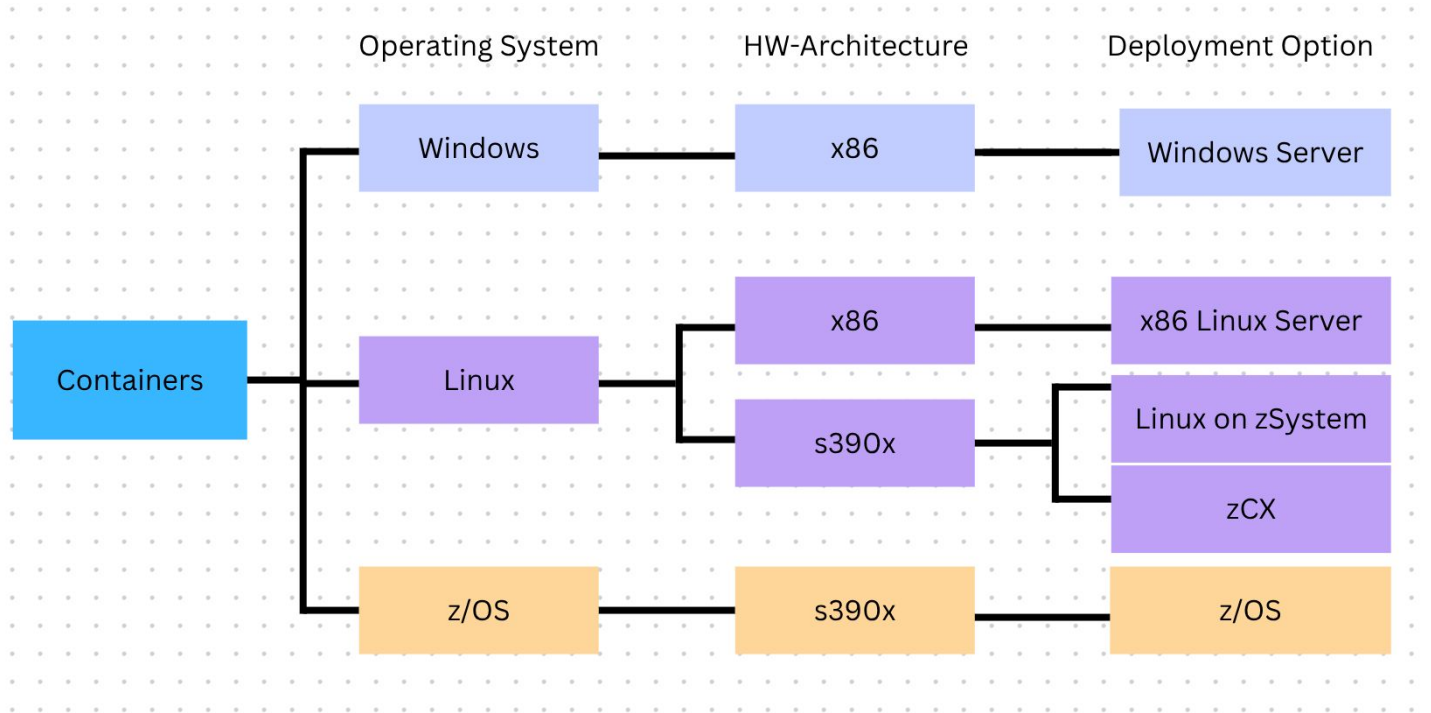


2. Why containers on Z?

1. It Works on My Laptop But Not on Z
2. Hard to access for modern developers
3. Running Multiple Workloads Without Conflict
4. Slow, Expensive Provisioning
5. Isolation and Security
6. Colocation with z/OS workloads
7. Address space communication
8. Inherit z/OS QOS



3. Types Container



3. Container options on Z

Linux on Z Containers

Packed Linux applications compiled for s390x hardware architecture for Linux on Z

Great for apps that want to leverage Z hardware capabilities

z/OS Container Extensions

Packed Linux application compiled for s390x hardware architecture for Linux on Z running inside of z/OS

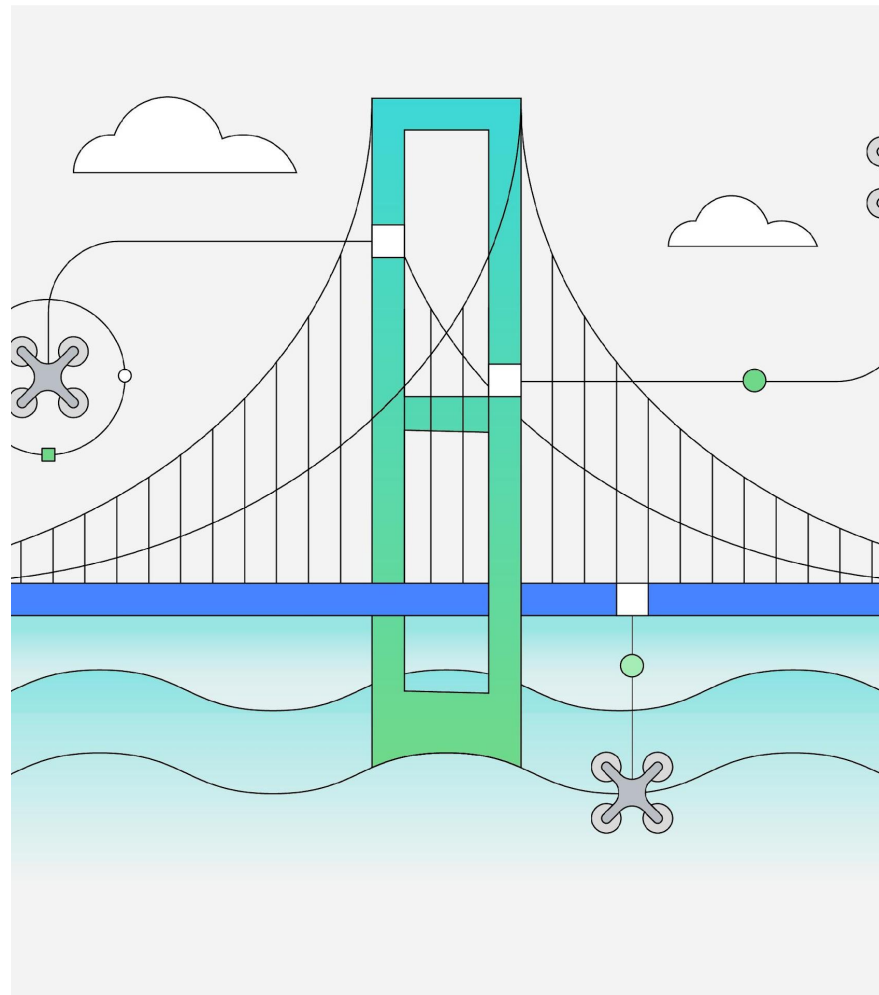
Great for linux applications that need access to z/OS applications

IBM z/OS Container Platform

Full enterprise Kubernetes cluster running on Linux LPARs, physically co-located with z/OS on the same Z hardware.



4. Torch-Spyre Profiling (Our Development)



4. AI workload on zCX